Aayush Paudel

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PROFESSIONAL SUMMARY

Data-focused student with experience building scalable data pipelines and performing analytical modeling on large datasets. Skilled in Python, SQL, and PySpark with strong interest in machine learning and statistical analysis.

EDUCATION

Bsc. CSIT **2023-Present**
St. Xavier's College Maitighar, Kathmandu

High School Diploma **2021-2023**
DAV College Jawalakhel, Lalitpur

PROJECTS

Nepal Power Grid Data Warehouse | Python, PostgreSQL, dbt, Airflow, Docker **Jan 2026 – Feb 2026**

- Built an end-to-end pipeline digitizing NEA's daily report PDFs into a star-schema data warehouse, extracting 33+ grid metrics (generation, imports, peak demand, system loss) using pdfplumber with BS↔AD date conversion and automated energy-balance validation
- Modeled dimensional transformations in dbt (staging → dimensions → facts) with derived analytics (import dependency %, season classification, fiscal year) and 100% pass rate across data quality tests
- Containerized the full stack with Docker Compose (PostgreSQL, Airflow, Metabase), enabling one-command deployment and daily automated scheduling via Airflow DAGs with idempotent upsert loading
- Set up CI/CD with GitHub Actions running pytest and full dbt builds on every push, validating 8+ automated test cases continuously

NEPSE Real-Time Streaming Pipeline | Python, Kafka, PySpark, TimescaleDB, Grafana, Docker **| Feb 2026 – Mar 2026**

- Engineered a high-throughput Python producer to simulate real-time market activity, replaying 12.5M historical NEPSE trades into a KRaft-based Apache Kafka cluster at 100x speed while guaranteeing strict event ordering by keying messages by stock symbol
- Developed a PySpark Structured Streaming application to consume high-velocity trade data, implementing 1-minute tumbling windows to continuously calculate OHLC (Open, High, Low, Close) candles, trade volumes, and VWAP metrics
- Containerized the event-driven architecture (Kafka, Spark, PostgreSQL, Grafana) using Docker Compose, configuring an optimized Spark JDBC sink to reliably land aggregated streaming data into TimescaleDB hypertables for time-series optimization
- Visualized live market liquidity and price movements via real-time Grafana dashboards, tuning Kafka producer configurations (batch.size, linger.ms, idempotence) to handle data skew and ensure exactly-once delivery semantics with <5s end-to-end latency

TECHNICAL SKILLS

Languages | Python, SQL
Frameworks & Tools | dbt, Apache Airflow, Apache Kafka (KRaft), PySpark (Structured Streaming)
Developer Tools | Git, Docker, GitHub Actions (CI/CD), Visual Studio Code, Make
Libraries | pandas, NumPy, Matplotlib, Seaborn, pdfplumber, psycopg2, PySpark SQL, confluent-kafka
Databases | PostgreSQL, Metabase (BI), Grafana, TimescaleDB